Musang Yoo

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Education

2022 – present **Ph.D. Candidate** in Geoinformatics, University of Seoul, Seoul, Korea

Advisor: Prof. Hyeongmo Koo

2017 – 2019 **M.S.** in Geoinformatics (Interdisciplinary Program), Kangwon National University at Gangneung, Gangneung, Korea
(M.S Thesis: Analyzing Characteristics of Migrant Households using Spatial Big Data)

Advisor: Prof. Chul Sohn

2010 – 2017 **B.A.** in Urban Planning and Real Estate, Kangwon National University at Gangneung, Gangneung, Korea

Professional Experiences

2019 – 2021 **Researcher,** Spatial and Environmental Research Division, Chungnam Institute, Korea

Research Experiences

2025 – 2026 **Principal Investigator,** Ph.D. Student Research Grant, Comparative analysis of varying local regression models according to spatial heterogeneity patterns, funded by the Ministry of Education, Korea

2022 – 2024 **Student Researcher,** Excellent Young Researcher Program, Developing machine learning methods incorporating spatial data properties, funded by the Ministry of Science and ICT, Korea

Teaching Experiences

2023. 03 – 2023. 06 Lab instructor and teaching assistant, Fundamentals of Geospatial Information Sciences, Dept. of Geoinformatics, University of Seoul, Seoul, Korea

Publications

- **Yoo, M.,** and Koo, H., 2027, Evaluating spatially varying coefficient models across structural patterns of spatial heterogeneity, *Geographical Analysis,* 59(1), e70070. https://doi.org/10.1111/gean.70070
- Koo, H., **Yoo, M*.,** Kang, H., Cho, Y., and Jeong, M., 2026, Evaluating the integration of spatial features in machine learning for model accuracy and residual autocorrelation, *Transactions in GIS,* 30(2), e70230. https://doi.org/10.1111/tgis.70230
- Cho, Y., **Yoo, M.,** Kang, H., and Koo, H., 2026, The impacts of scale effects of the modifiable areal unit problem on machine learning based urban development models, *Journal of the Korean Society of Surveying, Geodesy, Photogrammetry and Cartography,* 44(2), pp. 213-223. https://doi.org/10.7848/ksgpc.2026.44.2.213
- **Yoo, M.,** Kang, H., Cho, Y., Jeong, M., Koo, H., and Lim, S., 2025, Comparing local estimates of election vote share determinants using spatially varying coefficient models, *Journal of Korean Society for Geospatial Information Science,* 33(4), pp.89-107. http://dx.doi.org/10.7319/kogsis.2025.33.4.089
- **Yoo, M.,** and Koo, H., 2025, Exploring the impact of spatial autocorrelation on optimistic bias in cross validation and assessing the effectiveness of spatial cross-validation, *Cartography and Geographic Information Science,* 52(5), 596-609. https://doi.org/10.1080/15230406.2024.2422593
- Jeong, M., **Yoo, M.,** Kang, H., Cho, Y., and Koo, H., 2025, Impacts of spatio-temporal variation on estimation of spatial and spatio-temporal kriging, *Korean Journal of Remote Sensing,* 41(2), pp. 437-446. https://doi.org/10.7780/kjrs.2025.41.2.2.4
- Koo, H., **Yoo, M*.,** Seo, H., and Park, S., 2024, Exploring effectiveness of spatial features for incorporating spatial data into machine learning, *Journal of Korean Society of Surveying, Geodesy, Photogrammetry and Cartography,* 42(1), pp. 43-57. https://doi.org/10.7848/ksgpc.2024.42.1.1
- **Yoo, M.,** and Koo, H., 2024, Exploring discontinuous spatial heterogeneity in the determinants of apartment sales price in Seoul, *Journal of the Korean Cartographic Association,* 24(2), pp. 43-57. https://doi.org/10.16879/jkca.2024.24.2.043
- **Yoo, M.,** and Choi, D., 2019, An empirical study on the spatial effect of distribution patterns between small business and social-environmental factors, *Journal of Korean Association of Geographic Information Studies,* 22(1), pp. 01-18. https://doi.org/10.11108/kagis.2019.22.1.001
- **Yoo, M.,** Jeong, S., Kim, K., and Sohn, C., 2018, Seasonal analysis of beach-related issues using local newspaper articles and topic modeling, *Journal of Korean Regional Science Association,* 34(4), pp. 19-34. https://doi.org/10.22669/krsa.2018.34.4.019

Conference Presentations

- **Yoo, M.,** Kang, H., Cho, Y., and Koo, H., 2025, Comparing spatially varying coefficient models for election vote share determinants, *Conference of Korea Spatial Information Society* **(Best Presentation Award)**
- Jeong, M., **Yoo, M.,** Kang, H., Cho, Y., and Koo, H., 2025, Impacts of spatio-temporal variation sources on machine learning based spatio-temporal kriging, Korea, *Conference of Korea Spatial Information Society*
- **Yoo, M.,** Cho, Y., and Koo, H., 2024, Incorporating spatial data into machine learning using spatial features, *Conference of the Korean Association of Geographic Information Studies and the Korean*

Society of Surveying, Geodesy, Photogrammetry and Cartography **(Best Presentation Award)**

- **Yoo, M.,** and Koo, H., 2024, Exploring region-based discrete spatial heterogeneity in the determinants of apartment sale prices in Seoul, Korea, Korea, *Conference of the Korean Cartographic Association* **(Best Presentation Award)**
- Jeong, M., Kang, H., **Yoo, M.,** and Koo, H., 2023, Improving prediction accuracy through spatially constrained hierarchical clustering in machine learning models, Korea, *Conference of Korea Spatial Information Society*
- **Yoo, M.,** Park, S., Kang, H., and Koo, H., 2023, Exploring the effectiveness of the spatial features to incorporate spatial data into machine learning, Korea, *Conference of the GeoAI Data Society*
- **Yoo, M.,** and Koo, H., 2022, The effect of spatial autocorrelation on cross-validation and the validation of spatial cross-validation techniques, Korea, *Conference of the GeoAI Data Society*
- **Yoo, M.,** and Koo, H., 2022, Estimating the prediction performance of cross-validation in the presence of spatial autocorrelation, Korea, *Conference of the GeoAI Data Society*
- **Yoo, M.,** Jeong, S., Kim, K., and Sohn, C., 2018, Analyzing pending issues using topic modeling and local newspaper articles, Korea, *Conference of the Korean Regional Science Association*

Awards

- Outstanding Paper Award (Honors), University of Seoul, 2025. 02. 11.
- Outstanding Paper Award (Highest Honors), University of Seoul, 2026. 08. 13.

Research Interests

- Spatial Statistics
- Spatial heterogeneity and non-stationarity
- Machine learning
- Geographic Information System (GIS)